

ODRE PQC

Public Technical Whitepaper

ODRE PQC 0.3.1 RC1 · Release 63

Verified scope, authority, trust boundary, multi-platform delivery, FULL AUM and measured performance.

September 2026 · Release 63 technical baseline

1. Document authority and verdict

Release 63 is the Final Customer Delivery authority. The active Production pointer is Release 60; the two are intentionally distinct.

Authority	Verified state
Final Customer Delivery	Release 63 · g269 · 6/6 OS PASS
Active Production	Current 60 · Previous 59 · COMMITTED · Health PASS
Policy	Generation 1080 · Consumer Pin SYNCED
FULL AUM	g270 · A–R isolated PASS · destructive Production test NOT RUN BY POLICY

- Customer Delivery technical pass: YES; official supported OS: 6 of 6 PASS.
- Negative Gate: 18/18 PASS. Release 63, Sealed Core and g269 evidence remained byte-identical after closure.
- FULL AUM A–R was exercised with installed updater bytes and isolated fixtures; destructive Production injection was not run by policy.

Engineering interpretation

A later PASS does not erase an earlier FAIL. Every statement is limited to the artifact, platform, environment and gate identified by its evidence.

2. Product purpose

ODRE PQC binds post-quantum protected execution to signed artifacts, runtime identity, policy generation, consumer pin, license, lease and unit authority. A failed prerequisite does not downgrade to plaintext or classical fallback.

- Data Plane: Commercial Native and the approved Sealed Core perform protected operations.
- Control Plane: policy, consumer pin, identity and commercial entitlement determine whether execution is authorized.
- Evidence Plane: manifests, receipts, journals, canonical status and reports preserve the basis for PASS, FAIL and NOT RUN.

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3. Security architecture

Gateway, Native Runtime, Sealed Core, Policy, License and AUM are separate components but are joined by one authority at protected-request time.

- Missing or modified required artifacts fail before installation or apply.
- Policy and consumer pin mismatch produces zero protected execution and zero partial apply.
- Native/Core readiness or protected-request failure forbids commit.
- The Sealed Core is immutable, but it must be packaged, installed, loaded and exercised.

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4. Protected request contract

An authenticated request is accepted only after entitlement, identity, policy synchronization, Native readiness and Core loading are confirmed.

- The business handler is invoked exactly once for each measured successful protected request.
- Unauthorized execution, plaintext fallback and implicit classical downgrade are zero.
- A health code alone is not sufficient; the protected request is part of the commit gate.

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5. Release package trust chain

Selection is based on signed authority and exact SHA-256, never on the largest filename or most recent timestamp.

Generation	Role
60	Active Production baseline
61	Ubuntu runtime-manifest pin repair lineage
62	Immutable intermediate successor
63	Final Customer Delivery authority

- Product archive → signed inventory/manifest → envelope/CMS → installer binding → installed-byte load → protected integration → final ZIP seal.
- Release 60/61/62 bytes were not overwritten. Release 63 is a monotonic successor that closes the Ubuntu 24.04 Native authority lineage.
- Installer execution, installation integration, runtime integration and Customer Delivery PASS are separate gates.

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6. FULL AUM lifecycle

DISCOVER, VERIFY, STAGE, SWITCH, HEALTH, COMMIT, ROLLBACK and RECOVERY form a durable transaction.

- A single writer lock prevents concurrent apply transactions.
- Partially copied candidates are not promoted to VERIFIED.
- Atomic canonical status and a durable journal prevent torn JSON or a false COMMITTED state.
- Capacity is checked before staging; rollback assets are not deleted to make room.

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7. Services and administration

Windows services remain under the SCM lifecycle. start_server does not bypass the service contract by launching executables or Python entrypoints directly.

- Already-running services are not restarted; only stopped services are started through SCM.
- Admin Update is authenticated, dynamic and read-only.
- Observed Production: current 60, previous 59, floor 60, policy 1080, pin SYNCED, COMMITTED/PASS, health 8000/18443 HTTP 200.

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8. Official platform coverage

The same package, authority and protected-request contract passed on Windows Server 2022, Ubuntu 22.04, Ubuntu 24.04, Debian 12, Rocky Linux 9.8 and AlmaLinux 9.8.

OS	Result	Receipt SHA-256
Windows Server 2022	PASS	5E390E2062034FDE03BFA304D6A64EEAA191F79CB292E0EA913086CF5CB30489
Ubuntu 22.04	PASS	2E19C70F2992A1FA520E757DFC64A2FDB78DDEDBAB4E6AA4F943954D7B6E627D
Ubuntu 24.04	PASS	4AFBB78E6AB65D7D9389FA2B337C7E7350828AD69D827011CA399BAB226BC1B4
Debian 12	PASS	2597605241A07F53282EE0CEE6E5EF101919E178AAD00CBC06FF7FEBB2060293
Rocky Linux 9.8	PASS	701244BC819DA109AA190621F83EC91C36C5EDC0129041D80509F9AA952070AE
AlmaLinux 9.8	PASS	EBDBA21B870BD33032C26420669FC3F313890F45548B3369EB974C9E8FEB8563

- Windows-specific SCM/ACL and Linux service/permission adapters do not change the trust model.
- Containers, Kubernetes, other Windows versions and other Linux major versions are not claimed by this 6-OS verdict.

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9. Fail-closed and negative testing

Tampered ZIP/SHA, CMS failure, inventory mismatch, policy drift, duplicate/lower generation, runtime failure and insufficient capacity must stop before unsafe success.

- REJECTED_PRE_APPLY means Production change is zero.
- Post-switch health failure triggers restoration of Product, policy, pins and runtime authority.
- Historical FAIL evidence is retained after the minimal fix and successful retest.

Engineering interpretation

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10. Evidence and interpretation

g269 proves the Final Customer Delivery closure; g270 proves the bounded FULL AUM and live read-only Admin contract.

- Installed updater regression: 208/208 PASS; drop-in/existing AUM: 14/14 PASS; 12-A/Admin: 19/19 PASS.
- Drive upload/readback and two Notion history append/readbacks passed. Drive re-download SHA was not run by policy because no integrity anomaly was observed.
- No FIPS 140-3, Common Criteria, universal performance or universal platform claim is made.

Engineering interpretation

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Measured performance evidence

The figures below are from historical g108 Windows Server 2022 loopback candidate testing. They were not re-measured on Final Release 63 and must not be generalized to WAN or customer hardware.

Test	Measured value
Environment	Windows Server 2022 build 20348 · AMD EPYC 9454P · 4 cores · 16 GiB · loopback
Protected 1W max stable mean	1,176.9337 RPS (range 1,163.8582–1,190.0091)
Baseline 1W max stable mean	3,442.9391 RPS (range 3,413.4320–3,472.4462)
Same concurrency 2	Baseline 3,442.9391 / Protected 1,104.7263 RPS · reduction 67.9133%
Latency c2 baseline → protected	p50 0.5468→1.7223 ms · p95 0.7238→2.2881 ms · p99 0.9382→2.6597 ms
Protected security	200,128 success / 200,128 handler · invariant failure 0 · plaintext HTTP 426 · fallback NONE
Protected 2W / 4W	NOT_MEASURED · single_worker_lease_unavailable fail-closed
Baseline scaling reference	2W 5,035.1443 RPS c8 · 4W 5,475.0515 RPS c32 · zero error

- Same business_handler, POST /secure/echo and serialization were used for Baseline and Protected.
- The 2W/4W protected values were not estimated or multiplied.
- These figures do not support a claim that PQC overhead is negligible.

Conclusion

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Release 63 Final Customer Delivery is technically PASS. Active Production Release 60 remains a separate committed authority. Unexecuted scopes remain explicitly NOT RUN.